# Dhana Karthikeya Ventrapragada

+1 (669) 269-8955 | vdkarthikeya@berkeley.edu | linkedin.com/in/vdkarthikeya | github.com/vdkarthikeya | vdkarthikeya.github.io

## Education

**University of California, Berkeley** **Expected Dec 2027**

*Bachelor of Arts in Data Science and Applied Mathematics | GPA: 3.85/4.00* *Berkeley, CA*

- **Relevant Coursework:** Principles & Techniques of Data Science, Probability for Data Science, Introduction to Urban Data Analytics; *In progress:* Introduction to Machine Learning, Efficient Algorithms & Intractable Problems

## Technical Skills

**Languages**: Python, SQL, Java

**Product**: User Research, User Interviews, Discovery Read-outs, Documentation, Cross-Functional Collaboration

**Data & ML**: pandas, NumPy, scikit-learn, XGBoost, PyTorch, PostgreSQL, ArcGIS Online, Jupyter

**Engineering**: FastAPI, Supabase, Claude API, Apify, REST APIs, AWS S3, pytest, Git, GitHub

## Experience

**The After Collective | Fung Fellow, Health + Innovation | Berkeley, CA** **Aug 2026 - Present**

- Helped run **7** interviews with wildfire survivors and leaders (led **2**), finding insurance a shared struggle.
- Turned that into **3** AI-assisted concepts with teammates, took on a checklist tool for survivors, and gathered reactions.
- Took on testing the partner's recovery coach for gaps, and organize the team's documentation.

**Guang Labs | Engineering Intern | Remote** **May 2026 - Aug 2026**

- Built a service for an AI video startup that sorts videos and images into **6** content types, so the platform can handle each type differently.
- Added a confidence check that retries uncertain results on long videos, and stored each label with its confidence and reasoning so the team can audit it, backed by **86** tests.
- Developed an in-app notification bell that tells users when their video finishes processing or fails, instead of leaving them waiting on an idle screen.

**IDX Exchange | Data Science Intern | Remote** **Jun 2026 - Aug 2026**

- Contributed to a California home-price machine learning model trained on **71K** single-family sales; the final model's typical estimate was off by **8.6%** ($R^2$ of **0.88**).
- Traced an unusable first result to a data leak, fixed it, and brought accuracy back to a usable range.
- Measured accuracy by price tier and across **161** school districts, showing estimates were least reliable for luxury homes (**13%** error vs. **6.5%** mid-priced) and resort markets.

## Projects

**Hindsight (Cal Hacks 13.0)** | *Python, FastAPI, TypeScript, React, Supabase, Claude, Apify, Factory Droid SDK* GitHub

- Shaped the idea, features, and vision of a transfer planner for California community college students: code matches courses to official ASSIST agreements (alternatives, grades, agreement years) and plans terms around prerequisites.
- Claude reads confirmed transcripts and answers with code-checked sources; Apify scrapes campus pages and a Factory Droid agent repairs broken scrapers under deny-by-default permissions.

**Resume Tailor** | *Python, FastAPI, LaTeX, Claude* GitHub

- Built a tool that tailors a one-page LaTeX resume to a job posting, changing only the bullet text.
- Rejects rewrites that lack a quote from my own records, and checks one-page fit and ATS parsing.

**SightLine** | *Python, PyTorch, OpenCV* GitHub

- ML team member on a mobile app that turns a day of Meta Ray-Ban footage into a **60**-second recap video.

## Leadership & Involvement

**UC Berkeley CDSS | Dean's Cabinet Transfer Representative | Berkeley, CA** **Sep 2026 - Present**

- Advise CDSS leadership on the student experience for CDSS majors.

**UC Berkeley CDSS | Undergraduate Course Staff, CDSS 198 | Berkeley, CA** **Aug 2026 - Present**

- Give resume feedback in a careers seminar, and run exam reviews and practice worksheets for CS and data science classes for transfer students.

**UC Berkeley | Undergraduate Course Staff, CS 61A | Berkeley, CA** **Jun 2026 - Aug 2026**

- Taught **4** weekly exam-prep sections in Python, Scheme, and SQL, reinforcing data structures like linked lists, trees, lists, and dictionaries, held office hours, and helped write midterm and final questions.